SUPERSEDED FOR THE WIRING METHOD

On 8 October the builder chose bare brass rods instead of insulated wire: see the Brass Wiring page. The connections, tests and the layout on this page still apply; the wire table, lengths and the soldering-to-pins figure do not.

POCKET AI ASSISTANT · ENCLOSURE, STEP 4D · WIRING

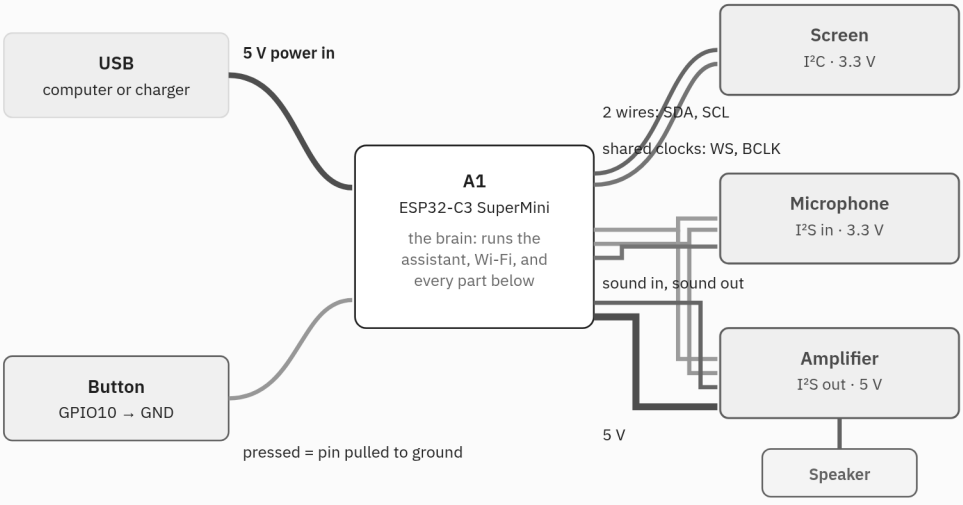
Wiring plan

How the parts talk to each other, every permanent wire, and how the whole thing fits inside the brass frame. You build the wiring on the bench this week, test it module by module, and drop the finished, working harness into the frame after it is soldered.

Where this fits

done	this week	Friday	after
Breadboard prototype	**Wire on the bench**	**Solder the frame**	**Into the frame**
everything worked on jumpers	permanent wires, off the frame, tested per module	Frame Soldering page	insulate, place, glue, Wi-Fi check

How the wiring works



One brain, four helpers. Coloured lines use the wire colours from the table below: yellow and white are the two clocks the mic and the amp share.

CONNECTION	WHAT IT DOES	WIRES
Power	USB brings 5 V into A1. A1 makes 3.3 V for the screen and the microphone, which must never see 5 V. The amplifier is the one part that takes 5 V straight from A1's `5V` pin. Every part shares one ground, the return path for all current.	5V, 3V3, GND
I²C (screen)	Two wires carry everything the screen shows: `SCL` is the clock, `SDA` the data. A1 calls the screen by its address, 0x3C or 0x3D.	2 + power
I²S (mic and amp)	Digital sound. Two clock wires tick 16,000 times a second for each sample: `WS` marks each sample and `BCLK` each bit. The mic and the amp share both clocks. Sound comes in from the mic on `SD` and goes out to the amp on `DIN`.	2 shared + 1 each way
Simple pins	The button pulls GPIO10 to ground when pressed. GPIO5 turns the amplifier on; its pull-down keeps the amp off while A1 starts up.	1 each
Speaker	The amp pushes and pulls the speaker with two outputs, `OUT+` and `OUT−`. Neither is ground, so they never touch GND or the frame.	2, twisted

This is exactly the circuit you already built and tested on the breadboard. The breadboard rails become short soldered wires and two small splices; nothing about the electronics changes.

Every connection

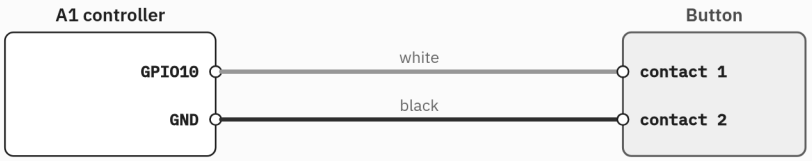
A1 PIN	GOES TO	COLOUR	WIRE	NOTE
5V	Amp VIN	red	26 AWG	the only 5 V wire; never to the screen or mic
GND	Amp GND	black	26 AWG	its own wire
GND	Splice → screen GND, mic GND, button	black	30 AWG	one wire to the pin, splice near the parts
3V3	Screen VCC	red	30 AWG	3.3 V only
3V3	Mic VDD	red	30 AWG	3.3 V only
GPIO20	Screen SCL	green	30 AWG	I²C clock
GPIO21	Screen SDA	blue	30 AWG	I²C data
GPIO1	Mic WS and amp LRC	yellow	30 AWG	shared word select: two wires on one pin
GPIO2	Mic SCK and amp BCLK	white	30 AWG	shared bit clock: two wires on one pin
GPIO4	Mic SD	green	30 AWG	sound coming in
GPIO3	Amp DIN	blue	30 AWG	sound going out
GPIO5	Amp SD	green	30 AWG	amplifier on / off
GPIO10	Button	white	30 AWG	pressed = to ground

> **COLOUR RULE**
>
> **black = ground**, **red = power**. The only **thick** red wire is 5 V and goes only to the amplifier. Signal colours repeat between modules, so put a small Kapton flag with the pin name on each end. If your wire kit's colours differ, keep black for ground and red for power and label the rest.

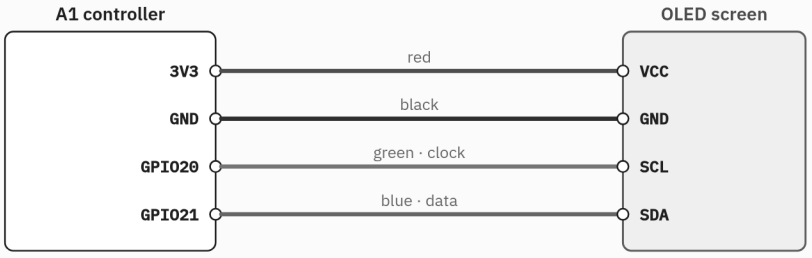
Small parts that live right at a module, not on A1:

AT	PART	BETWEEN	WHY
Mic	10 kΩ resistor	`SCK` and `VDD`	holds GPIO2, a start-up pin, high while A1 boots
Mic	100 kΩ resistor	`SD` and `GND`	keeps the data line from floating
Mic	short wire link	`L/R` and `GND`	selects the left channel the firmware reads
Amp	your `SD` pull-down (4.7 kΩ unless you chose 1 kΩ in Step 11)	`SD` and `GND`	amp stays off while A1 resets
Amp	100 kΩ resistor	`GAIN` and `GND`	the 15 dB volume setting you chose
Amp	220 µF electrolytic + 100 nF ceramic	`VIN` and `GND`	energy for loud sounds; stripe side to GND

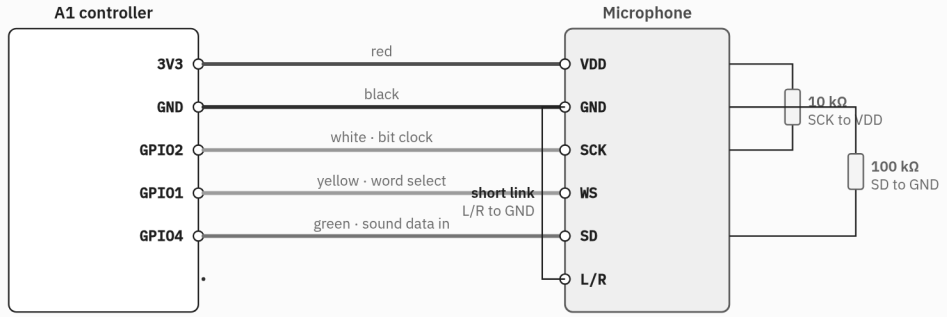
Module by module



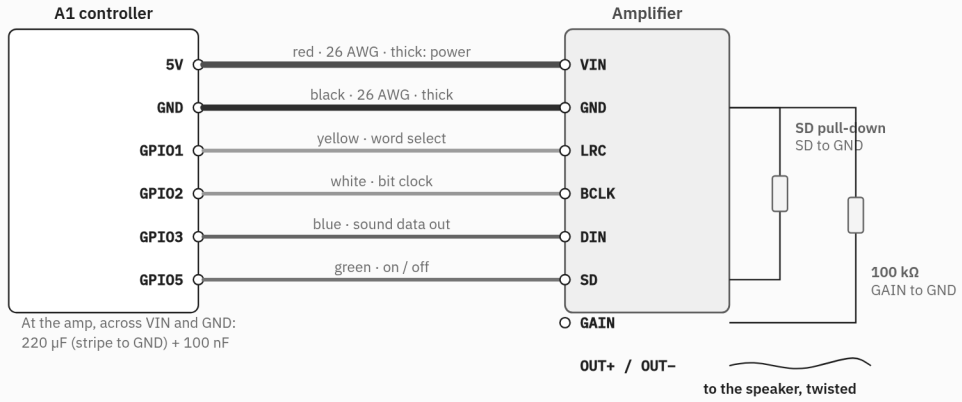
Two wires. Use the two contacts that you proved, with the meter, are open when released and joined when pressed. The firmware's internal pull-up holds GPIO10 high until you press.



Four wires, wired by the printed label. Identical-looking screens put GND and VCC in different orders, so read your board's labels, never a photo.

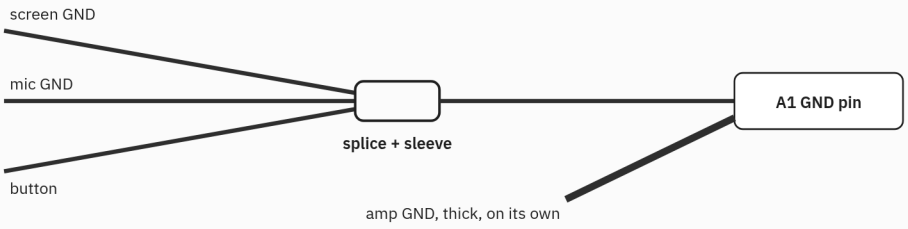


Five wires to A1, three small parts at the mic. L/R gets a short link to the mic's own GND pin (it picks the left channel). The 10 kΩ and 100 kΩ resistors you used on the breadboard now sit right on the mic's pins, which are the same electrical points as before.



Six wires to A1, four small parts at the amp, two leads to the speaker. The thick red wire is the only 5 V wire in the device and goes nowhere but VIN. Neither speaker output ever touches GND or the frame.

Shared pins and splices

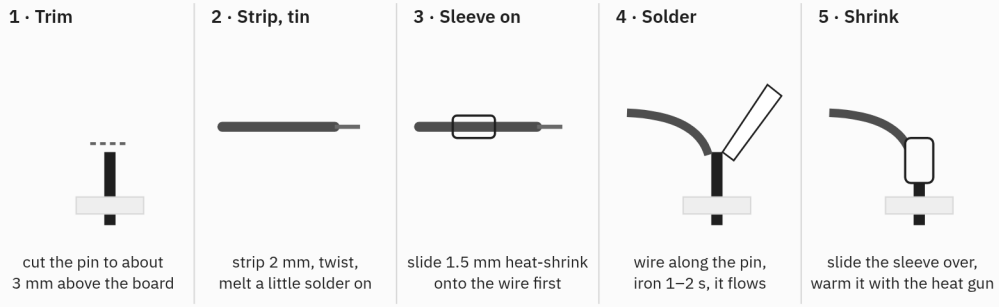


A1 has one GND pin and one 3V3 pin, but several parts need them. Join the quiet grounds in one sleeved splice with one wire to the pin. The amplifier's noisy, higher-current ground gets its own thick wire, so its switching never shares a wire with the microphone. The two 3.3 V wires (screen VCC, mic VDD) can both go on the 3V3 pin, or through a second small splice.

How to solder a wire to a header pin

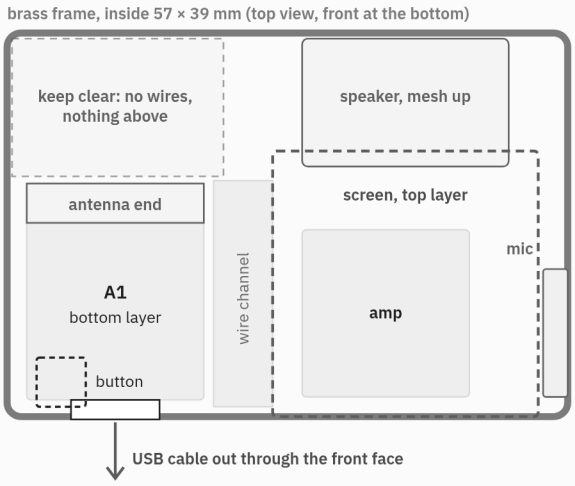
Use: the **electronics** tip, not the brass tip · Chip Quik no-clean flux pen, **never the Harris acid flux** · iron about 330–350 °C (625–660 °F) · USB unplugged

Your modules already have header pins from Step 5. Keep them and solder the wires to the pins: it is far less work than removing the pins, and the stack still fits with about 8 mm to spare.



Sleeve on before you solder. Once the wire is soldered you cannot get the heat-shrink over it any more. Every pin joint gets its own sleeve, so neighbouring pins can never touch.

How it fits inside the brass frame



Solid = bottom layer
Dashed = top layer

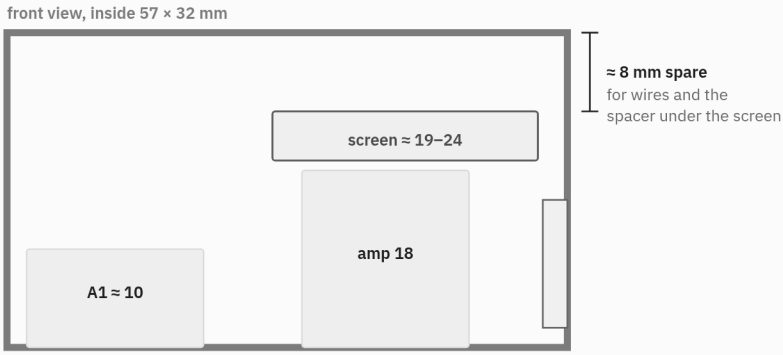
· A1's antenna points into the open back-left corner: about 15 mm to the back brass and nothing above it

· The screen stacks over the amp, not over A1

· The mic stands on the right wall, its port facing out, away from the speaker

· All wires run in the channel and along the floor, never across the antenna corner

Drawn to scale (7 px per mm), positions proposed. Part sizes are your caliper measurements; where each part sits is a first plan to confirm on the cardboard floor before any wire is cut.



Heights are estimates from your measurements: the amplifier's 18 mm includes its pins. A card or Kapton spacer sits between the amp and the screen's back so no pin touches another board.

> ONE CHANGE FROM YOUR MOCK-UP
>
> Your cardboard stack put the screen over A1. Its copper would then sit right over A1's antenna, which already needed its Wi-Fi power capped to be heard. This plan stacks the screen over the amplifier instead and leaves A1's antenna under open air, in the same 57 × 39 mm box. If you would rather keep the screen over A1, tell Claude: the wires stay the same, only some lengths change, and the Wi-Fi check after mounting decides.

Wire lengths

Lay the parts on a card cut to 57 × 39 mm in the positions above. Measure each wire's path with a piece of thread or the calipers, then **cut it 10 mm longer**: the spare lets you move a part when it goes into the frame. My first estimates:

WIRES	ESTIMATE	PATH
Button: 2 wires	≈ 30 mm	A1's side pins to the button over A1's USB end
Screen: 4 wires	≈ 45 mm	up the wire channel to the screen's pin edge
Amp: 6 wires	≈ 40 mm	across the channel; the two thick ones as short as possible
Mic: 5 wires	≈ 55 mm	along the floor under the amp to the right wall
Speaker: its own leads	do not cut	twist the pair, fold the spare length beside the speaker

Build order

> RULES FOR EVERY WIRING SESSION
>
> - **USB unplugged** for every solder joint and every change. Plug in only to test.
> - **Electronics tip and Chip Quik flux only.** The Harris acid flux and the brass tip never touch a module.
> - **Mic port:** no flux, heat, glue or air near its sound hole.
> - **Speaker leads:** never cut or strip them; they go into the amp's screw terminal.
> - **Read every label** on the board itself before soldering, especially the screen's VCC and GND.

1. **Lay out and measure.** Parts on the 57 × 39 card in the planned positions; measure and write down every wire length, plus 10 mm.
2. **Cut and prepare all wires**: strip 2 mm at each end with the Hakko stripper, twist, tin, and flag each end with its pin name.
3. **Button first** (2 wires). Test: plug USB in and press ten times; each press registers, as in fast-track Step 6.
4. **Screen** (4 wires). Unplugged meter check: VCC to GND gives no steady tone. Test: the screen lights at start-up and the log shows address 0x3C or 0x3D, as in Step 7.
5. **Microphone** (5 wires, the L/R link, two resistors). Test: speech makes the sound level rise compared with quiet, as in Step 8; on the assistant build, it hears a question.
6. **Amplifier** (6 wires, pull-down, GAIN resistor, two capacitors), speaker *not* yet connected. Meter, unplugged: 5V to VIN continuous, VIN to GND no steady tone, OUT+ and OUT− not connected to GND. Powered: SD below 0.08 V before speech, as in PASS 11A.
7. **Speaker on**, unplugged, into the amp's terminal. Test: five spoken replies in a row, clear, no reset, nothing warm, at cap 45, as in PASS 11C.
8. **Two cold starts** with everything wired. Then the harness is done.

- [ ] **WIRE-01 · THE HARNESS WORKS OFF THE FRAME**
  All five modules pass their tests on the permanent wires, two cold starts are clean, and every joint has its sleeve. Tell Claude, and send a photo of the harness laid out on the card.

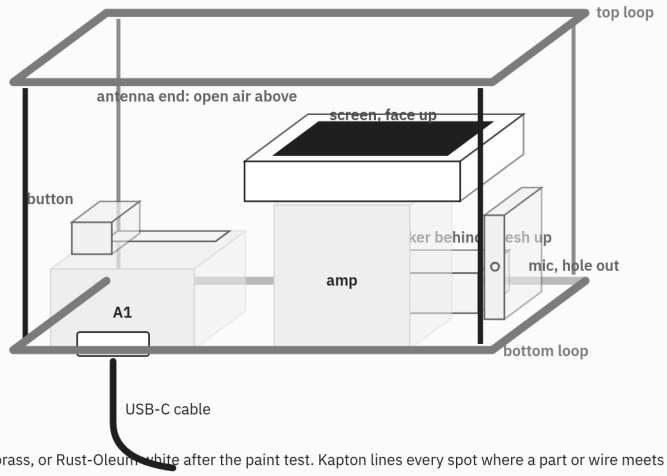
Claude can read the test logs for you: plug A1 in, say which module you just wired, and Claude opens the monitor and checks the result.

Then into the frame

1. **FRAME-01 done**, washed and dry (Frame Soldering page).
2. **Project rules updated** to allow the isolated frame. Claude makes that change with your OK.
3. **Kapton** on every module back and wherever a wire or pin could touch brass.
4. **Place the harness** in the frame, nothing glued, and re-run the full test.
5. **Fix the modules** with dabs of low-temperature hot glue, away from the mic port and connectors.
6. **Wi-Fi check**: compare signal and dropouts inside the frame with the open bench.

The finished device

see-through view from the front, a little above · to scale, 6 px per mm · outside 60 × 42 × 35 mm



Raw brass, or Rust-Oleum white after the paint test. Kapton lines every spot where a part or wire meets brass.

What you are building. A see-through brass cage the size of a matchbox: screen reading up, speaker grille up at the back, one USB-C cable out of the front. Every opening the cardboard mock-up asked for is here: USB at the front, the open top is the screen window and the speaker grille, and the mic hole and button face the sides.

Component check

Every part in the plan, checked against the hardware contract and the fast-track steps you passed:

PART	CONNECTION AND SETTINGS	WIRES	CHECKED AGAINST
A1 · ESP32-C3 SuperMini	flashed, PASS 10C; 8.5 dBm Wi-Fi cap	5V, 3V3, GND, GPIO1, 2, 3, 4, 5, 10, 20, 21	✔ matches the hardware contract in CLAUDE.md
OLED SSD1306 128×64	I²C on GPIO21 SDA / GPIO20 SCL, address 0x3C or 0x3D	4 wires	✔ Step 7
INMP441 microphone	I²S in: WS GPIO1, SCK GPIO2, SD GPIO4; L/R → GND = left slot; 10 kΩ up on GPIO2, 100 kΩ down on GPIO4	5 wires + link + 2 resistors	✔ Step 8; nothing on GPIO8
MAX98357A amplifier	I²S out: LRC GPIO1, BCLK GPIO2, DIN GPIO3; SD GPIO5 with pull-down; GAIN 100 kΩ to GND = 15 dB; VIN from 5V only; 220 µF + 100 nF	6 wires + 4 parts	✔ Step 11 and the 2026-09-30 GAIN change
Speaker CES-20134-088PM, 8 Ω 0.8 W	OUT+ / OUT−, never GND; volume cap 45 (2.20 V peak worst case < 2.53 V)	its own 2 leads	✔ PASS 11C at cap 45
Button	GPIO10 to GND, internal pull-up	2 wires	✔ Step 6; GPIO9 BOOT stays free
Brass frame	isolated: no current, no GND, no OUT±; Kapton at every contact; antenna end open	0 wires	⚠ allowed only once the project rules are updated with your OK
Battery, charger, regulator, switch	not in this device	none	✔ out of scope by your decision of 2026-09-21

Pin count on A1: 5V ×1, GND ×2 (amp, splice), 3V3 ×2, GPIO1 ×2, GPIO2 ×2, GPIO3, GPIO4, GPIO5, GPIO10, GPIO20, GPIO21 ×1 each: **16 wire ends on 11 pins**, all on the two header rows you already soldered.

Your kit

PRODUCT YOU BOUGHT	USE	LINK
Meshnology ESP32-C3 SuperMini (A1)	the controller	B0F888JQ91
Hosyond SSD1306 OLED, 0.96″	screen	B09T6SJBV5
AITRIP INMP441 microphone	mic	B092HWW4RS
HiLetgo MAX98357A amplifier	amp	B0CDWXZZCH
Same Sky CES-20134-088PM speaker	speaker	DigiKey CES-20134-088PM
QTEATAK tactile button	button	B0FHW6HMG4
CBAZY 30 AWG silicone wire, 6 colours	all signal and 3.3 V wires	B073RDGTPB
TUOFENG 26 AWG silicone wire	the amp's 5V and GND	B07G2LRX68
LuminologyPro resistor kit	10 kΩ, 100 kΩ ×2, SD pull-down	B0F4P352BB
ALLECIN electrolytic and BOJACK ceramic kits	220 µF and 100 nF at the amp	owned · no link recorded
Pointool heat-shrink kit, 14 sizes, white	a sleeve on every joint and splice	B08N4W4K9X
QWORK mini heat gun	shrinking the sleeves	owned · no link recorded
Hakko CSP-30-1 wire stripper	stripping 2 mm cleanly	B00FZPHMUG
MAIYUM 63/37 solder, 0.8 mm	every joint	B076QF1Y85
Chip Quik CQ4LF no-clean flux pen	electronics flux	owned (Adafruit order) · no link recorded
JoTownCand solder wick	clearing a bridge	B0DRN688Q5
ELEGOO Kapton tape	wire flags now, insulation later	B072Z92QZ2
KAIWEETS HT118A multimeter	continuity and voltage checks	B08BL288LW
NEIKO digital caliper	measuring wire paths	B000GSLKIW
Amazon Basics cardstock	the 57 × 39 layout card	B0DSH5V1TT

Sources. Every connection, resistor and capacitor comes from `plan/FAST_TRACK.md` Steps 6–11 and the hardware contract in `CLAUDE.md`; the 100 kΩ GAIN resistor is the 15 dB setting from the Louder Speaker page. Part sizes are your caliper measurements of 28 September; the frame size is from the frame template. Product links are from `plan/FAST_TRACK_PARTS.md`, `docs/INVENTORY.md` and the original order sheet.

What is and is not verified. The circuit is the one you tested on the breadboard. The layout inside the frame, the heights and the wire lengths are first estimates, not yet checked on parts. The checkbox is remembered only in this browser.